

"Through the Solar System" ——XR science education system based on multiple monitors

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ABSTRACT

This paper presents a multi-user extended reality (XR) science education system based on multiple monitors. It effectively uses existing equipment and space, allowing teachers and students to share immersive virtual scenes, increasing the possibility of collaboration, and attracting students through interactive interfaces and plot designs. Actively learn scientific knowledge, face-to-face communication, and interaction, thereby enhancing the effect of science education.

Keywords: Extended reality, multiple monitors, science education.

Index Terms: Human computer interaction—Extended reality

1 INTRODUCTION

Science education with immersive media technology can transform students from books into highly interactive digital experiences. Studies have shown that applying science education can promote students' enjoyment of learning [6], promoting knowledge retention and interaction through experience models [1]. However, there is little practice due to limited cost control. This paper reviews previous related work, then describe the XR system design and take "Through the solar system"(TSS) to show the system implementation process in specific topics. Finally, we present some preliminary user studies' evaluation results.

2 RELATED WORK

There are currently some widely used systems, like head-mounted display systems (HDS), CAVE, and mobile device systems. Users of HDS have the highest sense of immersion. Still, the dizziness and heavy equipment are inevitable, the isolation of reality will also cause users to perform some dangerous activities [4] and neglect to break the boundaries between people. The CAVE simulates a panoramic experience by setting up three or more LED screens in the space. The view is broad, allows users to maintain their perception of each other and the shared physical environment [2]. However, the larger space and expensive display devices are not conducive to popularization. The popularity of smartphones provides convenience and low cost for mobile device systems, but the primary and secondary schools do not allow children to use smartphones in the classroom. The interactive and collaborative forms of touch screens are also limited.

Research by Cliburn et al. pointed out that part of the lack of VR applications is high-cost teaching equipment [3]. Freitas and Campos [5] show that students' use of shared displays can increase classroom collaboration. Inlian Radu proposed real-time

operation requires the stability of the wireless network [7]. The multiple monitors system presented in this paper has an advantage in cost control, because multi-screen exists in modern classrooms or low-cost purchases. The local connection between the virtual space and the actual display avoids the network's impact on the experience. Of course, compared with the head-mounted display and CAVE system, the XR system's user immersion based on multiple displays has limitations. However, we think it can become a bridge between expensive virtual reality technology in science education and normal video playback on the Internet.

3 SYSTEM DESIGN

People who look out through the windows of cars, spaceships, submarines, and so on have a broad view, so we designed a new system between the cockpit system and the CAVE system. The system uses multiple monitors to simulate windows and display real-time photos taken by cameras hanging on the virtual space carrier. Students can experience the carrier's movement but stay in a classroom safely and interact with each other.

According to the research of Cliburn et al., the three-screen system has the highest efficiency among multi-screen display systems [3], so we set three monitors, which are on the front, left, and right sides of the room. The carrier model we build will be proportional to the physical space's size to enhance the sense of reality. The position of the camera and the real-space display will be consistent. The user will also use a host to simulate a carrier's console, such as a simulated car driving position, and control various carrier movements through an input device to achieve picture changes, as shown in Figure 1.

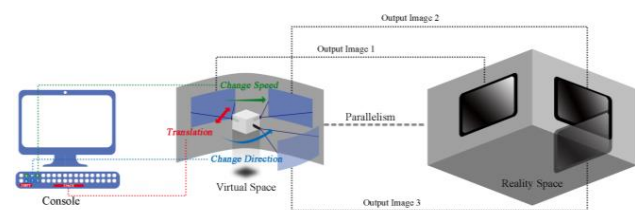


Figure 1: Controllable carrier output camera picture on display.

The system is also equipped with background music to increase the auditory experience and novice teaching to help teachers and students understand different topics' operational content. During the use of the system, the order in which users watch the screen is random because we hope that users can explore different sides to watch, which can make them feel that they are inside the carrier and see the disappearance and reappearance of certain scenes with the movement of the carrier.

4 SYSTEM IMPLEMENTATION

We designed the XR system TSS with the theme of representative astronomy knowledge in science education for elementary and middle school students.

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4.1 TSS Design and Implementation

We use Unity3D to build virtual scenes of the solar system, include a scale model of the sun, eight planets, and the moon. Then, we set them to simulate rotation and revolution based on planetary trajectories, speed, and inclination data. The system carrier is a spacecraft in TSS, and the target display channels connect the images recorded by the three cameras. We set up a host to connect to three displays simultaneously, and the live image of the corresponding channel is output, as shown in Figure 2. When the spacecraft moves, the image on display will also change continuously. The user can probably see a planet disappearing from the left screen and appearing on the front screen, and finally disappearing on the right screen.



Figure 2: The final effect picture and real scene picture.

4.2 Interaction in TSS

The host of TSS simulates the "driver's station" of a spacecraft. The users use the keyboard to control spacecraft's movement, include forward and backward, four steering lines, and speed control. We set up a knowledge introduction module according to the content requirements of science education. When the users operate the spacecraft to reach the designated location, they will trigger the event button. Both sides of the screen will display the planet's text introduction, and then spacecraft follows the planet motion according to the pre-set follow-up script. The students can read the text introduction while observing the planet's real-time motion images to understand scientific knowledge more intuitively and vividly. In this time, the users can still manipulate the front camera's rotation script to rotate within the view field's limit to facilitate observation.

We display the spacecraft's real-time position coordinates and the radar map in the corner to promote collaboration and interaction. The teacher can organize that some students judge the direction based on the macro position and tell the other students who control the "driver's station" to operate, promoting students to express and cooperate. Teachers can also ask questions directly, such as finding the position of Jupiter, comparing the size of Jupiter and the earth, etc. students discuss and perform the operating routes they planned, then try to answer the questions through observation. This process will promote interactive collaboration in the classroom and enhances students' space, imagination, organization ability, etc.

5 USERS FEEDBACK

Due to COVID-19 restrictions, we only conducted a small pilot study in the conference room that possessed three monitors to evaluate TSS usability. We invited two elementary school science class teachers and eight students (average age=11) to participate in the test, and the ratio of males to females was the same. We trained the teachers on the system operation in advance, observed the course's interactive content during the process, and organized interviews with the subjects at the end.

We interviewed teachers and students separately, and the questions for students are simple, including whether they liked the system or which process they impressed deeply. Most students(six) are excited in TSS and actively experienced the "driver's station." They said, "this interactive form is more than my previous courses,

and I have the feeling of moving fast," "Look closer the sun, and it is much bigger than I thought," the other two students feel a little down, they mentioned, "It's not easy to drive a spacecraft well." "When I encounter introduction parts, I control less than others." When interviewing teachers, we focused on assessing the system's usability and its impact on the course. The two teachers generally recognized this system, and they said, "I think the initial use of the system is not complicated, but it sets higher requirements for us to prepare lessons because of uncertain order of knowledge." "The system provides students to look at the specific process when I talk about it. They will participate in control the courses by their operation so that they will be more active than I control the courses." But they also put forward, "To ensure immersion, we must close the curtains." Some parts of the system needed tweaking from the current feedback, but the overall review was positive. Participants also expressed expectations for developing more themes of the XR science education system.

6 CONCLUSION AND FUTURE WORK

Based on the universal monitors that exist or low-cost purchases as the hardware foundation, we carry on XR's system design, which strikes a balance between cost and immersion and reduces space and network demand. Connecting virtual space and reality through screens creates a stronger sense of space than video playback. Sharing screens enhances the possibility of collaboration, interactive design is more attractive to students, and enrich the teaching methods. Next, we consider doing formal user research about the system's adaptive, controllability, and impact on knowledge retention. We hope that cheap hardware and easy-to-use interaction design can make more people have excited and valuable science education XR experience.

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